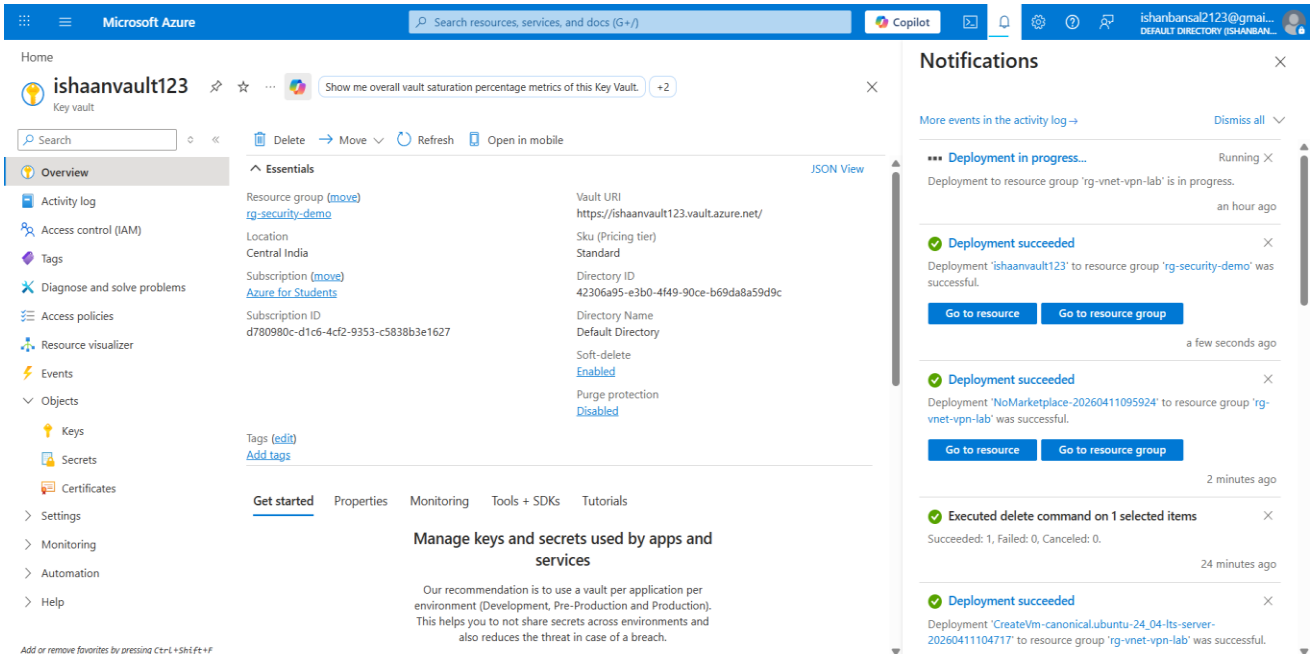
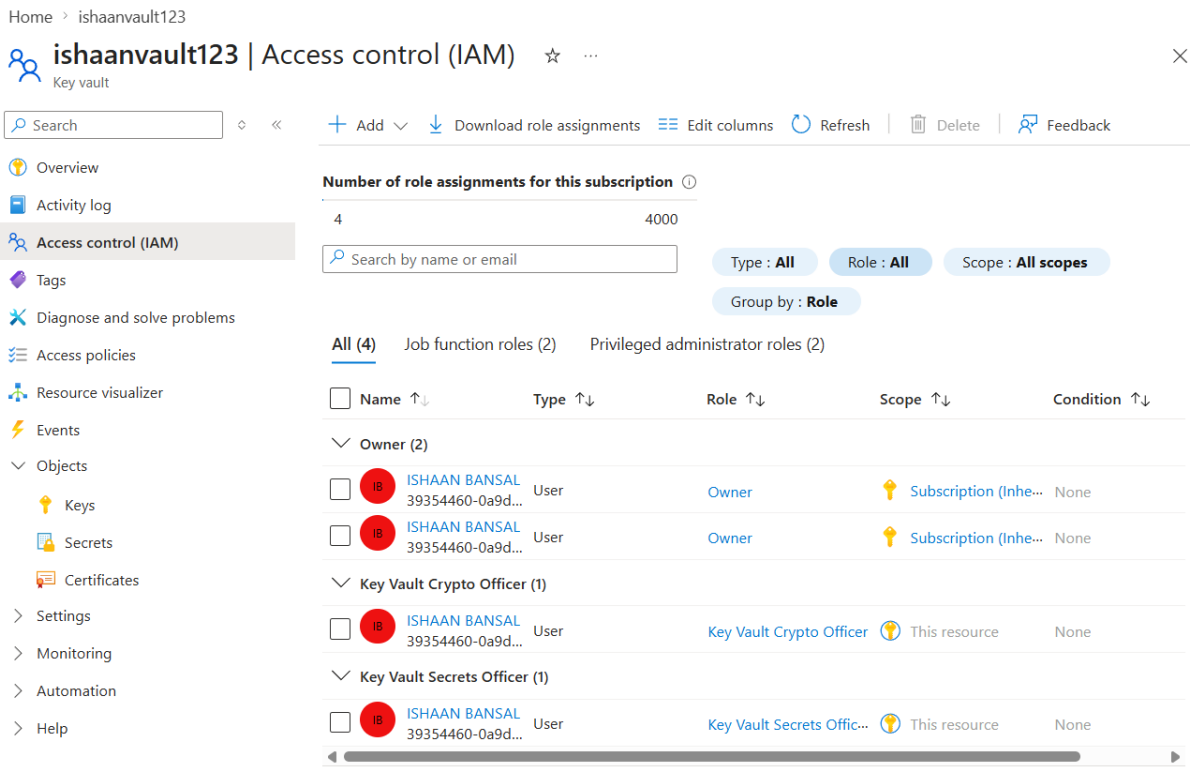
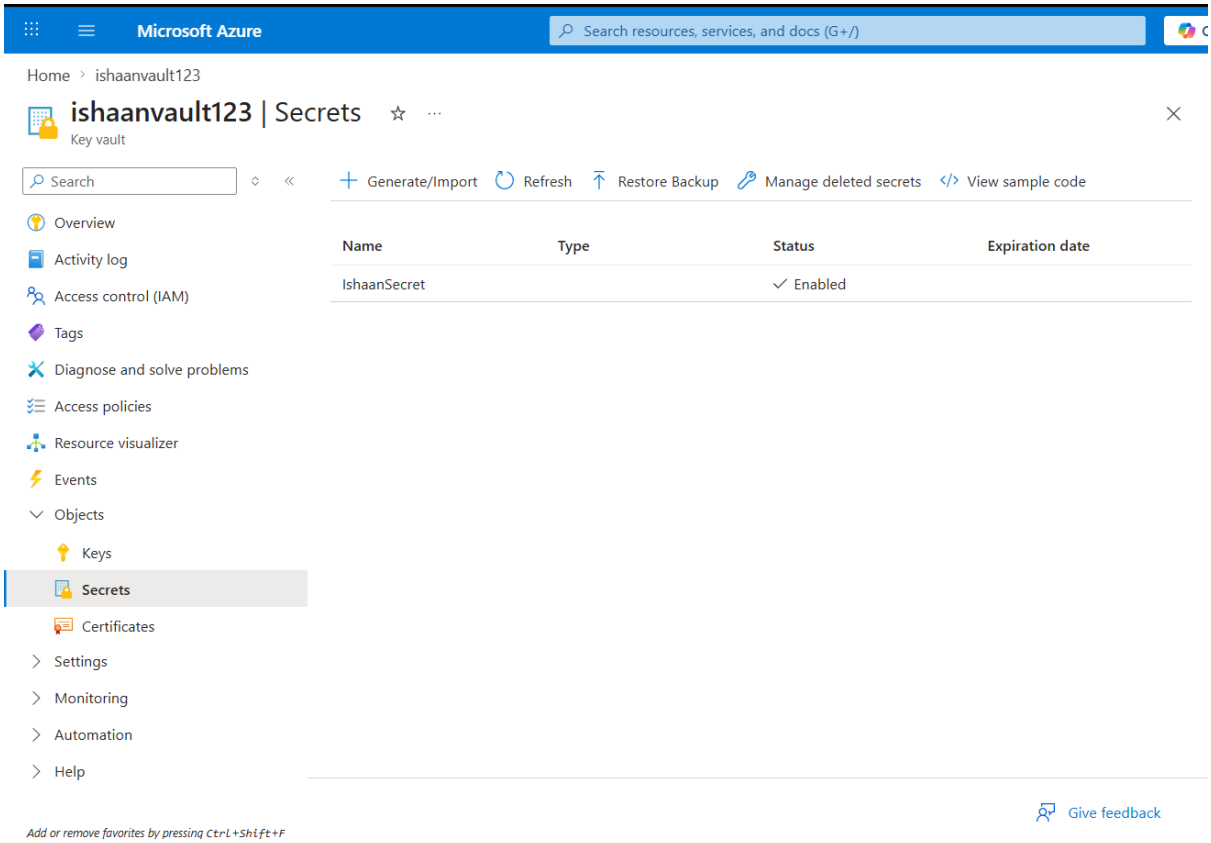
1. Create Key Vault



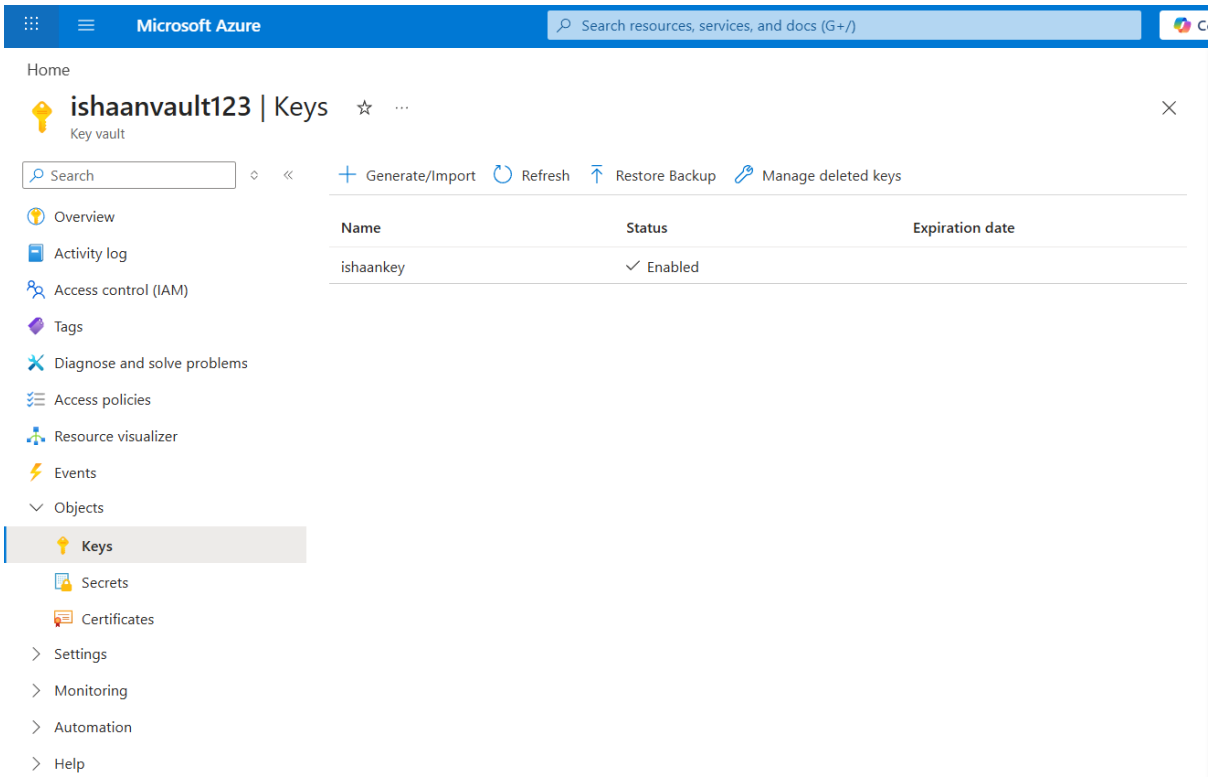
2. Giving Access



3. Creating Secret



4. Creating Key



5. Checking it in Console App

```
internal class Program
{
    0 references
    static void Main(string[] args)
    {
        string keyVaultUrl = "https://ishaanvault123.vault.azure.net/";
        string keyName = "ishaankey";

        var credential = new DefaultAzureCredential();

        var cryptoClient = new CryptographyClient(
            new Uri($"{keyVaultUrl}keys/{keyName}"),
            credential
        );

        string plainText = "Hello SmartBank!";
        byte[] data = Encoding.UTF8.GetBytes(plainText);

        // Encrypt
        var encryptResult = cryptoClient.Encrypt(
            EncryptionAlgorithm.RsaOaep,
            data
        );

        Console.WriteLine($"Encrypted: {Convert.ToBase64String(encryptResult.Ciphertext)}");

        // Decrypt
        var decryptResult = cryptoClient.Decrypt(
            EncryptionAlgorithm.RsaOaep,
            encryptResult.Ciphertext
        );

        string decryptedText = Encoding.UTF8.GetString(decryptResult.Plaintext);
        Console.WriteLine($"Decrypted: {decryptedText}");

    }
}
```

6. Console Output

```
Microsoft Visual Studio Debu

Encrypted: i9I5ZUONP8VIrRaCK9rJb7vC7n+D9Ex91h5tJhGBuFl6UfNRp/wyZSYzT1Y0fPCIA2eHH5Mh+KOFRCcpNXRHxXRM0VVC7bmovwqmKSOP0AXi7
GY3JE5lu3XPwb1419d38fPG6twuZ0eKy2nd/O7uppbQxqI7EoCaN4ZKd4a2sJYgrcRlVgbgAla1Vhg6AtSOl5pK4+x12lDrPSQfywIAbmmaUQhGXejM9umgt
MMjwTRGW/KV0++XUTYHAW0SmWN7ghmrF9yQoHdcLvZXfxqu5Yd4veIz2Sv2wHVqmRiJXOurtirQQ/CxU/+sLC9IgwXfELgkplsvTIoLx6jckgqQYA==
Decrypted: Hello SmartBank!

D:\DotNetTrainingExtended\ConsoleAzureDEMO\ConsoleAzureDEMO\bin\Debug\net8.0\ConsoleAzureDEMO.exe (process 23072) exited
 with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```